

Methodology for the detection of numerical anomalies in the use of BEM solvers

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ABSTRACT: The boundary element method (BEM) solvers WAMIT, NEMOH, and HAMS share similar computational frameworks and solution workflows, each relying on dedicated geometry, input, and configuration files. These tools can be efficiently executed via command-line interfaces, which are used to analyse the hydrodynamics of floating bodies within diverse marine industry applications. Despite their methodological similarities, each solver exhibits distinct numerical characteristics, capabilities, and limitations that may affect accuracy and computational efficiency. This paper presents a methodology for obtaining hydrodynamic coefficients using these widely adopted BEM codes, establishing a comparative analysis that considers their three main operational stages: pre-processing, processing, and post-processing. This iterative method can be employed in performance studies and for identifying numerical errors, primarily associated with the wetted surface integration for computing coefficients in both the frequency and time domains.

Keywords: WAMIT, NEMOH, HAMS, Iterative method, numerical issues, time domain

Table – TD coefficients comparison by similarity criterion between solvers by $\Delta\omega_k$ variation

		Kr_{33}	Kr_{44}	Kr_{55}	$Ke_3 (0^\circ)$	$Ke_4 (45^\circ)$	$Ke_5 (90^\circ)$
$\Delta\omega_1$	S_1	1.000	1.000	1.000	1.000	0.999	0.936
	S_2	0.994	0.989	0.993	0.967	0.994	0.947
	S_3	0.986	0.988	0.983	0.957	0.958	-6.692
$\Delta\omega_2$	S_1	1.000	1.000	1.000	1.000	0.999	0.937
	S_2	0.994	0.990	0.993	0.978	0.994	0.947
	S_3	0.986	0.989	0.983	0.984	0.959	-6.400
$\Delta\omega_3$	S_1	1.000	1.000	1.000	1.000	0.999	0.938
	S_2	0.994	0.990	0.993	0.994	0.994	0.950
	S_3	0.986	0.989	0.983	0.997	0.959	-6.601

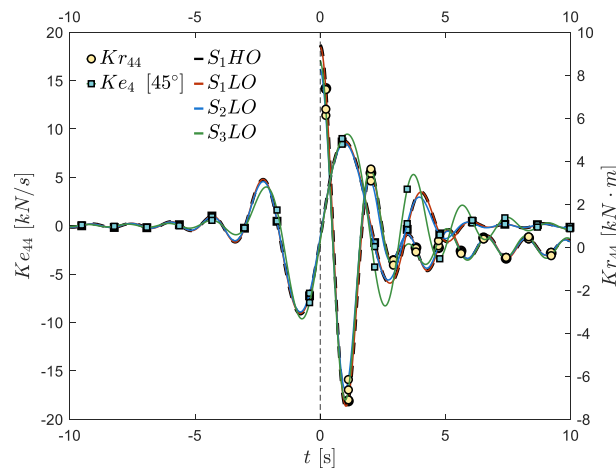


Figure - TD Hydrodynamic coefficients Kr, Ke in DOF [3,4,5] to meshing P_2 with time increment Δt_3 . S_2 with variable frequency spacing $\Delta\omega_k$ to heave motion